Loans vs Subsidies: Lithuanian State Support Policies During the COVID-19 Pandemic —Preliminary: Do Not Cite Without Permission—

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The views expressed here are those of authors and do not necessarily reflect the view of the Bank of Lithuania

Introduction

- Covid-19 pandemic $\rightarrow$ unprecedented economic shock
- Governments intervened by rolling out policies for firms and households
- What we know? Implications of government support for firms and households (Chetty et al., 2021, Autor et al., 2022, ...)
- What we don't know? Heterogeneous impact of heterogeneous policies
- This paper:
  - Builds unique administrative data for Lithuania: a combination of state support data with firms' characteristics:
    - Firm balance sheet information
    - Employment at the quarterly level, turnover at the monthly level
  - Set of stylized facts on state support policies for firms in Lithuania
  - Employment growth of loans versus subsidies receiving firms

Data Sources and Construction

- KOTIS: Suteiktos Valstybės Pagalbos ir Nereikšmingos (De Minimis) Pagalbos Registras (`kotis.kt.gov.lt`)
  - Administered by the Competition Council of the Republic of Lithuania
  - Detailed information on support provided by state to legal entities in Lithuania
  - Information contains: unique receiver ID, date, type, amount of support, etc.
  - Web-scrapped all information in September 2022
- Data on employment subsidies: used to be available on the Employment agency's website, collected and shared by the BoL staff
  - Total subsidies during downtime and total subsidies after downtime
  - No disbursement dates
  - Some amounts are non-disclosed

Data Sources and Construction

- Tarpžinybinė Duomenu Saugykla (TDS): the registry data for the universe of *formal* firms in Lithuania Reason for non-uniformity of data
  - Information contains: employment, firm's balance sheet, monthly turnover based on VAT declarations, NACE industry classification, etc.
  - Not homogenized across years
  - Present analysis uses data from 2015Q1 to 2023Q1
- Merge datasets using unique firm identifiers
- Sample selection:
  - Drop budgetary institutions, associations, non-profit organizations, and financial firms
  - Drop firms with omitted information on employment
- Aggregation: subsidies, grants + tax exemptions, loans + tax deferrals

Lithuanian State Support Policies for Firms

Roll out of policies

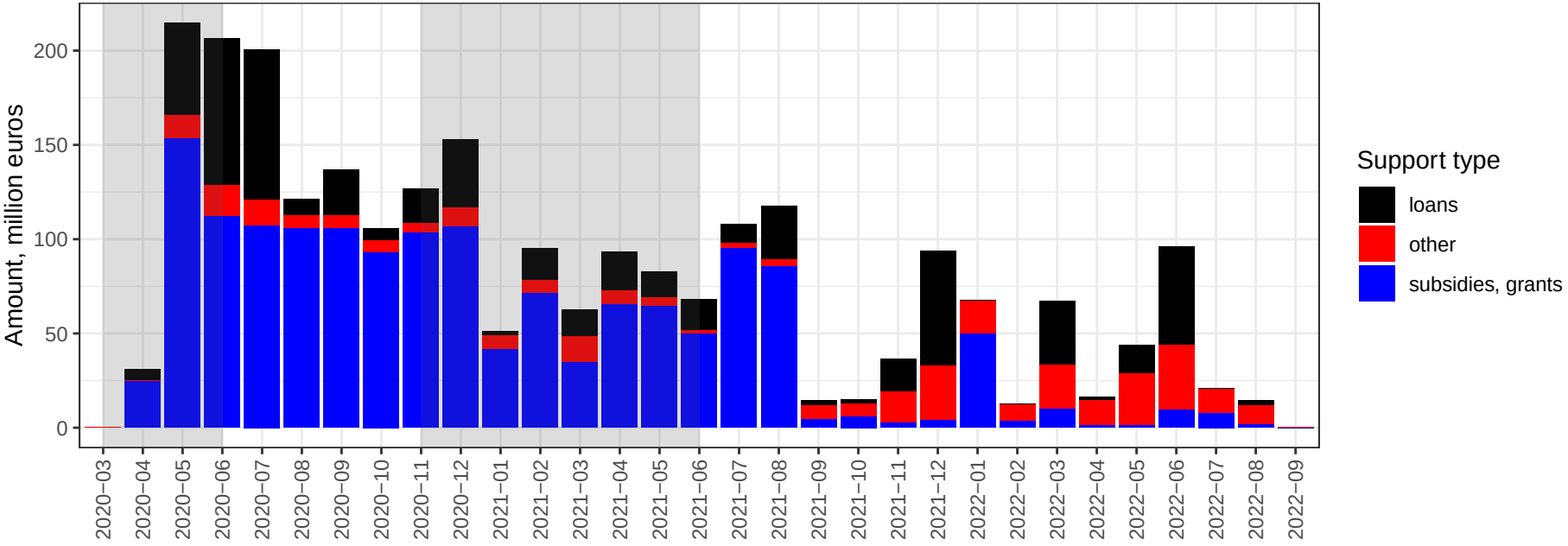
- Roll out occurred in two phases:
  1. Phase 1: March 2020+: initial phase, uncertain environment, broad range of measures, weaker eligibility criteria
  2. Phase 2: November 2020+: more targeted measures, stricter eligibility criteria
- Administration of support through the involvement of governmental ministries and agencies
  - Ministry of Economics and Innovation introduced initiatives to enhance businesses' liquidity
    - Executed by Investment and Business Guarantees LTD. (INVEGA) and the State Tax Inspectorate (VMI)
  - Ministry of Finance and VMI facilitated tax deferrals and tax exemptions
  - Some support measures also involved municipal governments

COVID-19 Support by Provider and Support Type (in Thousand Euros)

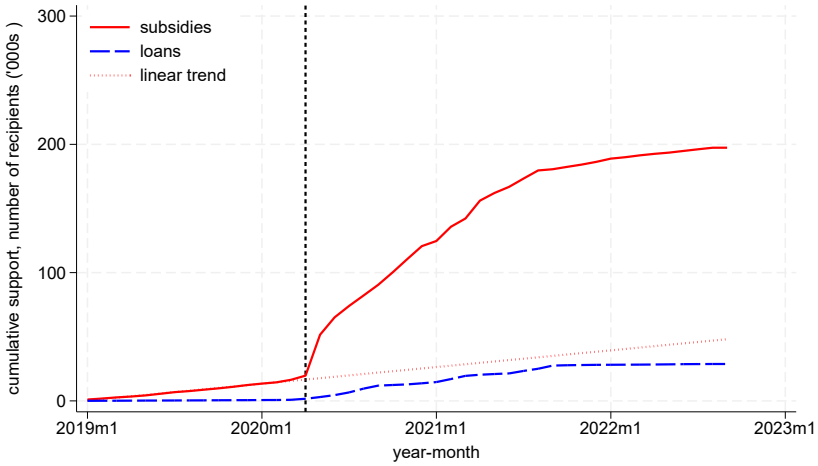
Type of support:	guarantee (1)	loan (2)	subsidy, grant (3)	tax deferral (4)	tax exemption (5)	total (6)
Ministry of Social Security and Labor			997,650			997,650
Ministry of Economics and Innovation	74,567	322,265	472,943			869,775
VIVA		224,568				224,568
Ministry of Finance	213,014					213,014
ACGF		72,680	1,548			74,228
VMI				44,392		44,392
Ministry of Culture			13,527			13,527
Ministry of Agriculture		50	12,641		6	12,697
Ministry of Energy			9,893			9,893
other			7,872		12	7,885
SODRA	6	0		5,581		5,587
NMA			4,471			4,471
total	287,587	619,568	1,520,548	49,993	35	2,477,731
% total support	12	25	61	2	0	100

Source: KOTIS Database (2020–2022).
Description: This table compiles the total amounts of COVID-19 support, classified by support provider (or administrator if the implementing and administrating agencies differ) and type from 2020 to 2022. We define COVID-19 support as outlined in the main text, with the classification of support types given in KOTIS.

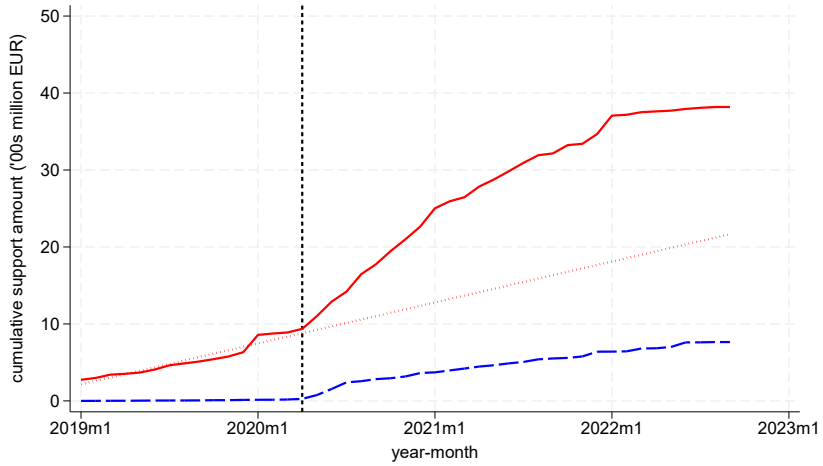
Timeline of disbursement of COVID-19 support



The number of recipients



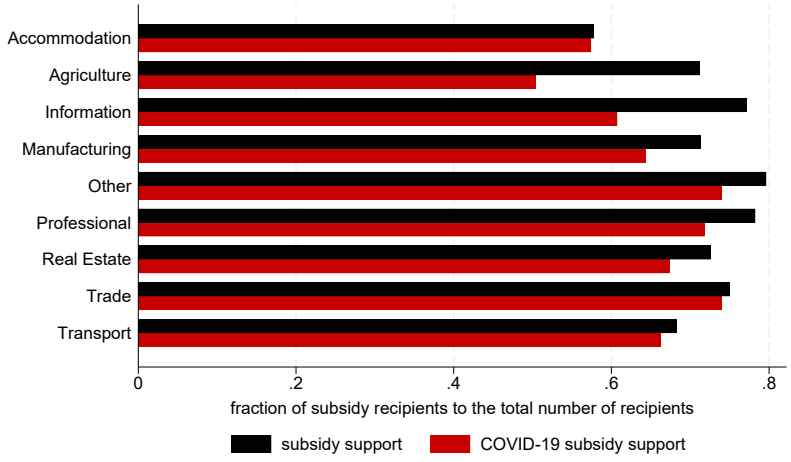
(a) Cumulative number of recipients



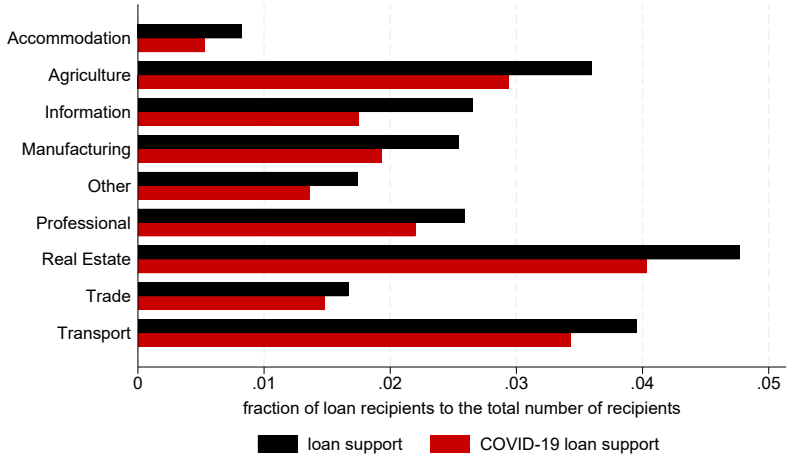
(b) Cumulative disbursed amount

- Loans and subsidies are two largest categories (in terms of total amount)
- However, the number of loan applications is low
- Our aggregation: subsidies, grants + tax exemptions, loans + tax deferrals

State support conditional on industry and the type of support

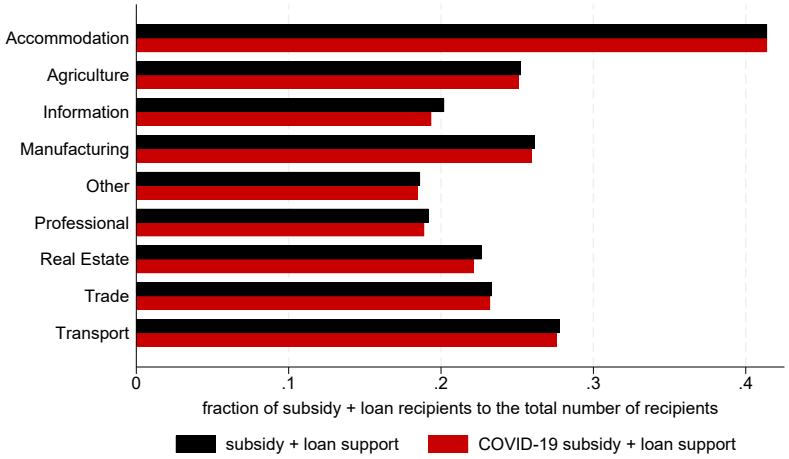


(a) Subsidy support only



(b) Loan support only

State support conditional on industry and the type of support

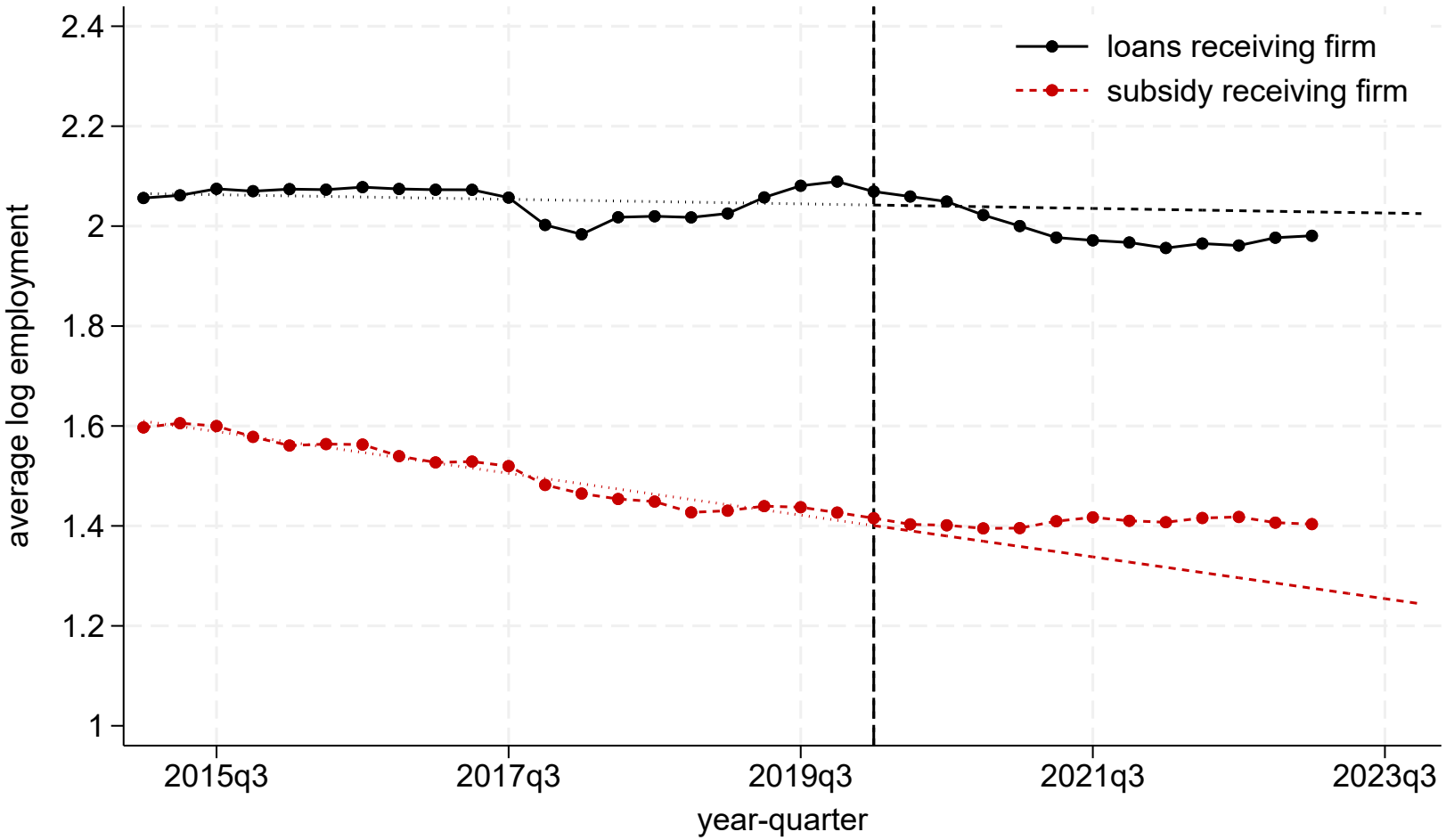


(a) Subsidy+ loan support

- Small but non-negligible fraction of firms received only loans
- Is the choice between loans and subsidies innocuous?
- We look through the lens of employment (growth)

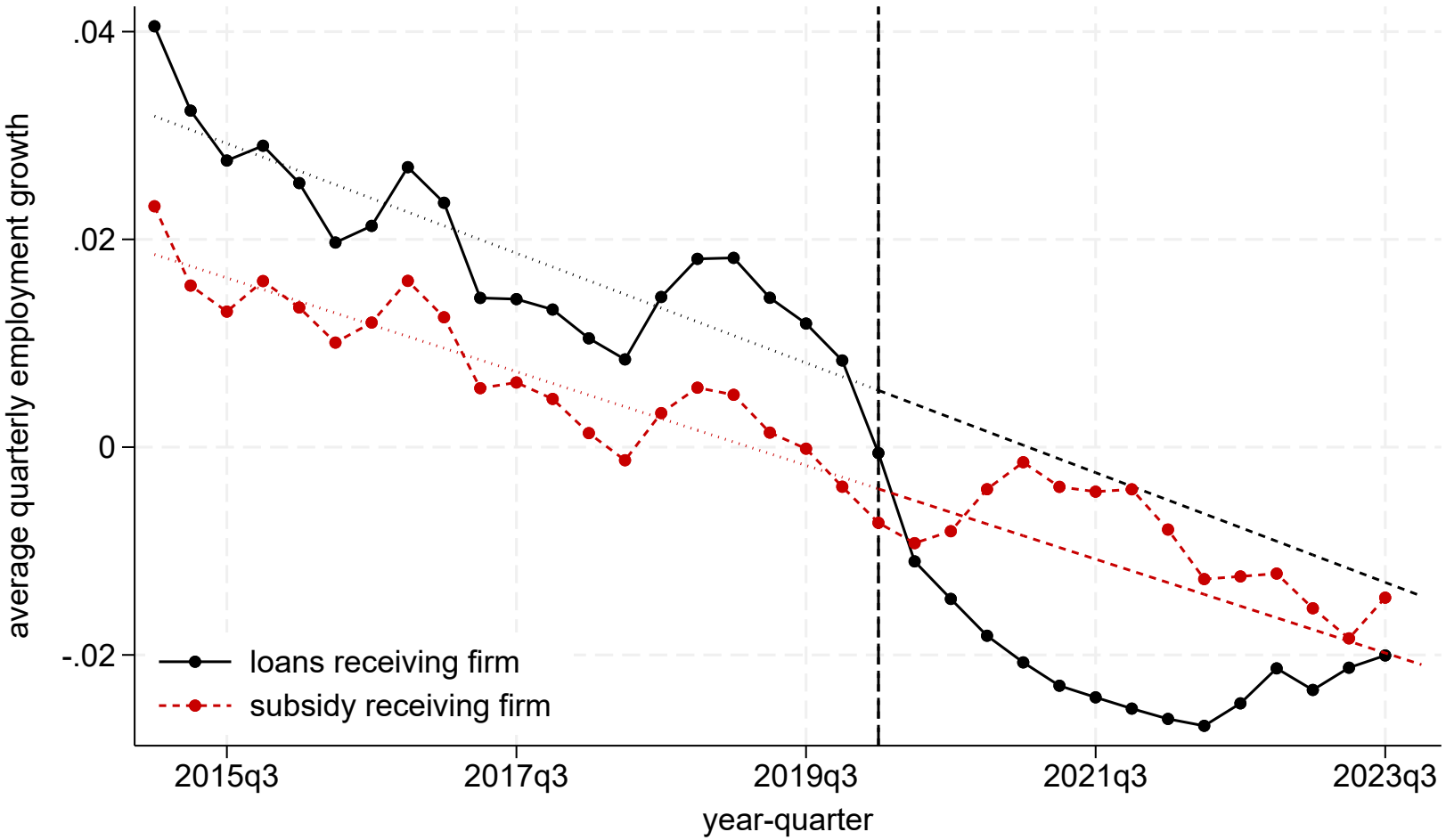
Employment of Loans- vs Subsidies-Receiving Firms

Employment level



Only Covid marked support

Employment growth



Only Covid marked support

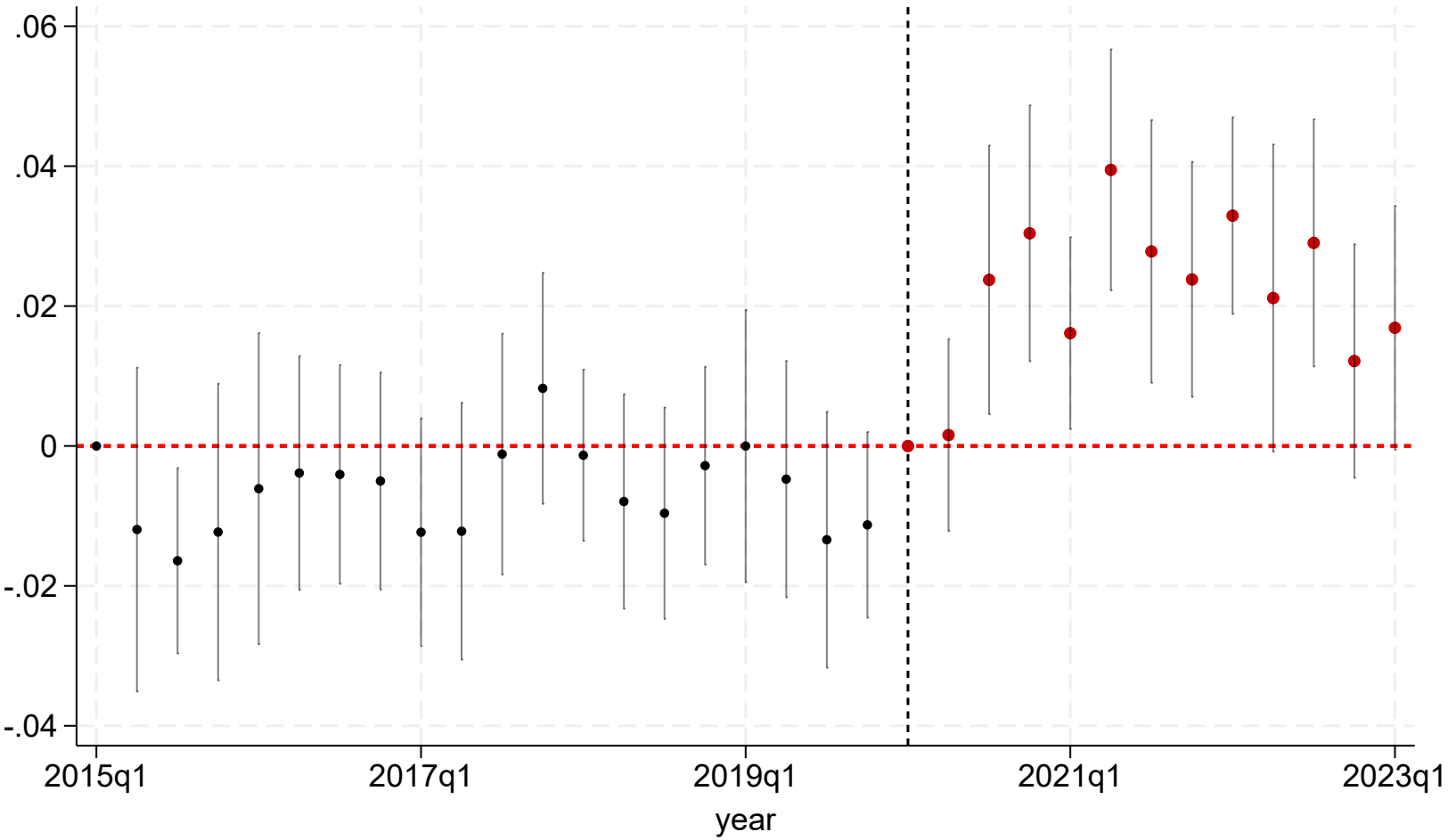
Empirical specification

- We run the following difference-in-differences specification:

$$\Delta L_{idt} = \beta_0 + \beta_1 s_i + \sum_{r=2015,Q2}^{2019,Q4} \beta_r \cdot (s_i \times \gamma_r) + \sum_{r=2020,Q2}^{2023,Q1} \beta_r \cdot (s_i \times \gamma_r)$$
$$+ \alpha_t + \alpha_i + \alpha_{dt} + \varepsilon_{idt}$$

- Notation:
  - $i \rightarrow$ firm, $d \rightarrow$ industry, $t \rightarrow$ year-quarter
  - $L_{idt} \rightarrow$ log employment, $s_i \rightarrow$ subsidy receiving firm
  - $\alpha_t, \alpha_i, \alpha_{dt}$: year-quarter, firm, and industry-year-quarter fixed effects
- Cluster standard errors at the industry-year-quarter level

Relative employment growth of subsidy-receiving firms



COVID-19 support | Excluding employment subsidies | Excluding taxes

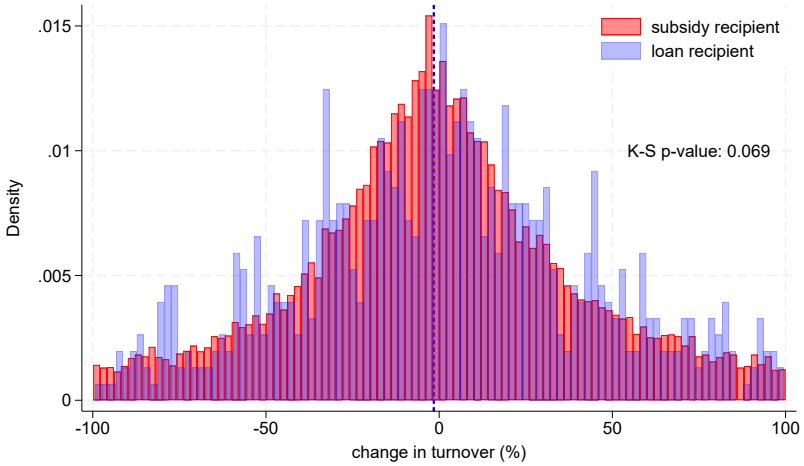
What explains the choice between taking a loan or subsidy?

- Three factors can explain this:
  1. Eligibility criteria
  2. Severity of the pandemic shock
  3. Self-selection criteria

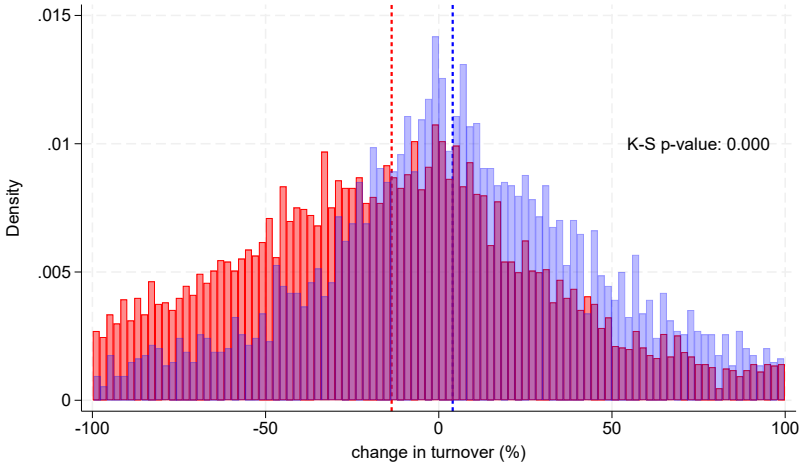
1. Eligibility criteria

	Micro	S	M	L	Income fall >= x	Quarantine-affected	VMI list	Planned budget (million euros)
Direct COVID-19 loans (phase I)	Y	Y	Y	Y	Y	Y		36
Loans to businesses through alternative financers "Alternatyva"	Y	Y	Y					50
Direct COVID-19 loans (phase II): business expenses	Y	Y	Y	Y		Y		100
Direct COVID-19 loans (phase II): investment	Y	Y	Y			Y		100
Loans for large and medium-sized businesses			Y	Y				100
Loans to ensure liquidity of agricultural and fisheries companies	Y	Y	Y	Y	Y			160
Loans for businesses most affected by COVID-19	Y	Y	Y		Y			200
Partial compensation for rent expenses for companies most affected by COVID-19	Y	Y	Y			Y		28
Compensations for employees' COVID-19 testing	Y	Y	Y					30
COVID-19 R&D	Y	Y	Y	Y				30
COVID-19 products LT	Y	Y	Y					31
E. commerce model COVID-19	Y	Y	Y					40
Compensation for interest rate payments	Y	Y	Y					57
Subsidies for micro companies	Y						Y	100
Subsidies for businesses most affected by COVID-19	Y	Y	Y	Y	Y	Y		120
Subsidies for businesses afffected by COVID-19	Y	Y	Y		Y			150
Subsidies during downtime	Y	Y	Y	Y				473
Subsidies after downtime	Y	Y	Y	Y			Y	473

2. Severity of the pandemic shock



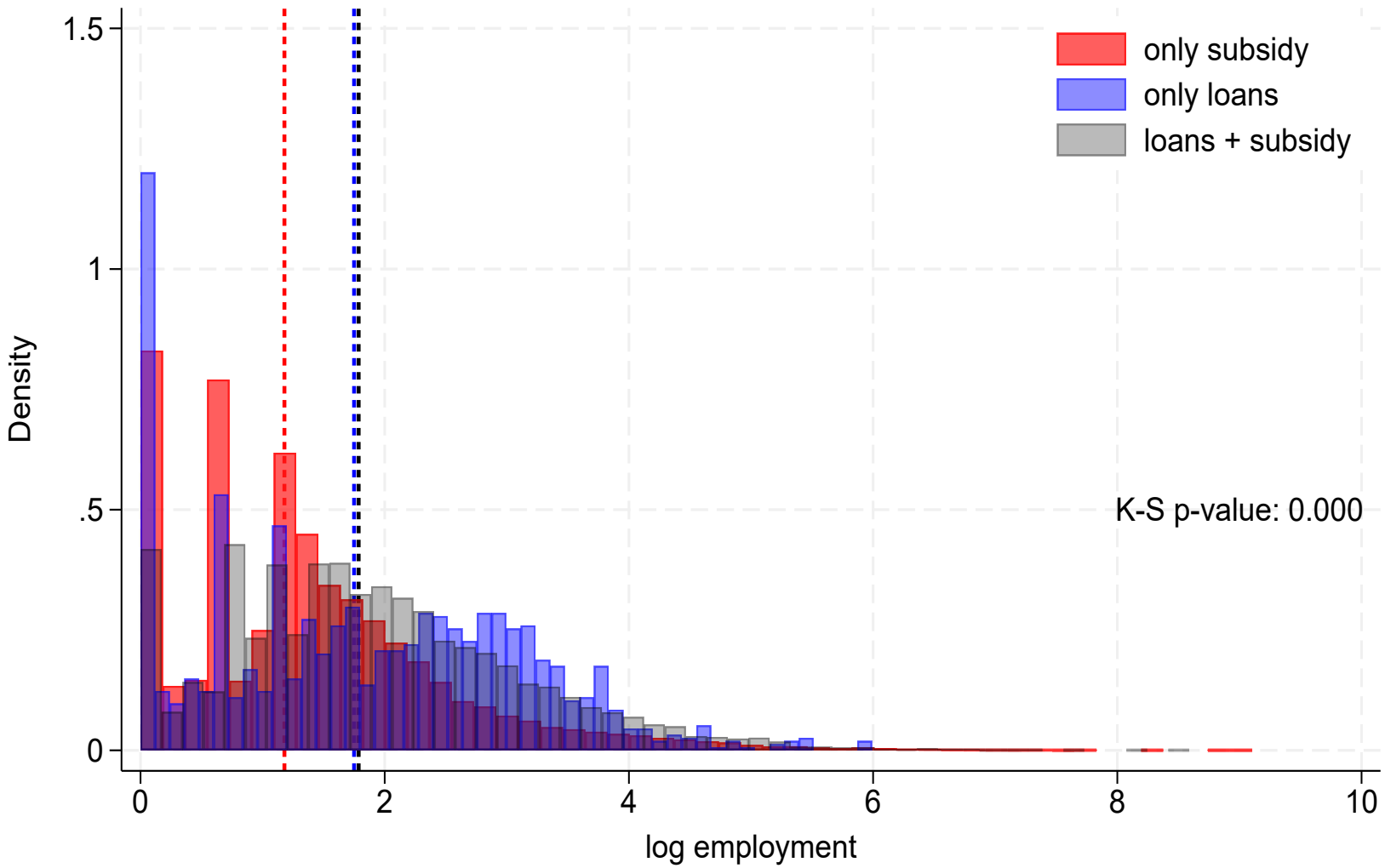
(a) 2019Q1–2020Q1, all support recipients



(b) 2019Q1–2020Q1, supported before November, 2020

3. Self-selection criteria

- Underlying characteristics of firms led to the choice of loans over subsidies
- Main factor: firm size



3. Self-selection criteria

Dep. var.:	Type of support indicator			
Base outcome:	No support			Subsidies only
Other outcome(s):	subsidies only (1)	subsidies + loans (2)	loans only (3)	loans only (4)
log employment	0.394***	0.729***	0.679***	0.270***
	(0.011)	(0.016)	(0.042)	(0.045)
log total assets	-0.138***	-0.036*	0.072	0.257***
	(0.014)	(0.021)	(0.063)	(0.070)
log current assets ratio	-0.026***	-0.060***	0.089**	0.153***
	(0.010)	(0.015)	(0.045)	(0.049)
log labor productivity	0.207***	0.249***	0.245***	0.051
	(0.007)	(0.011)	(0.033)	(0.036)
Observations	85,942			34,354

SMEs | Quarantine-Affected Firms | Revenue Fall $\geq$ 30% | Quarantine-Affected Firms + Excluding Taxes | Revenue Fall $\geq$ 30% + Excluding Taxes

Conclusion and Going forward

- We bring two administrative datasets together to study the impact of state support policies on firms in Lithuania
- We find that firms receiving only subsidies and tax exemptions exhibit relative higher employment growth during the pandemic
- This effect is driven by persistent negative employment growth of firms receiving loans and tax deferrals
- We find that self-selection criteria play a role for the choice between loans and subsidies even after accounting for policy eligibility criteria or severity of the pandemic shock
- Going forward: Model to explain why firms choose loans over subsidies
- Tentative hypothesis: Firms taking loans want freedom to choose how to adjust their labor
- If true, will be visible in the firm's productivity growth

Reasons for the non-uniformity of the TDS data

- Three reasons:
  1. All formal firms are required to submit their financial reports to SECR, some firms can submit subset of information and are exempted from submitting profit/loss and cash flow statements
  2. Required content in financial reports change over time due to changes in the underlying law
  3. Although required by law, non-compliance is present

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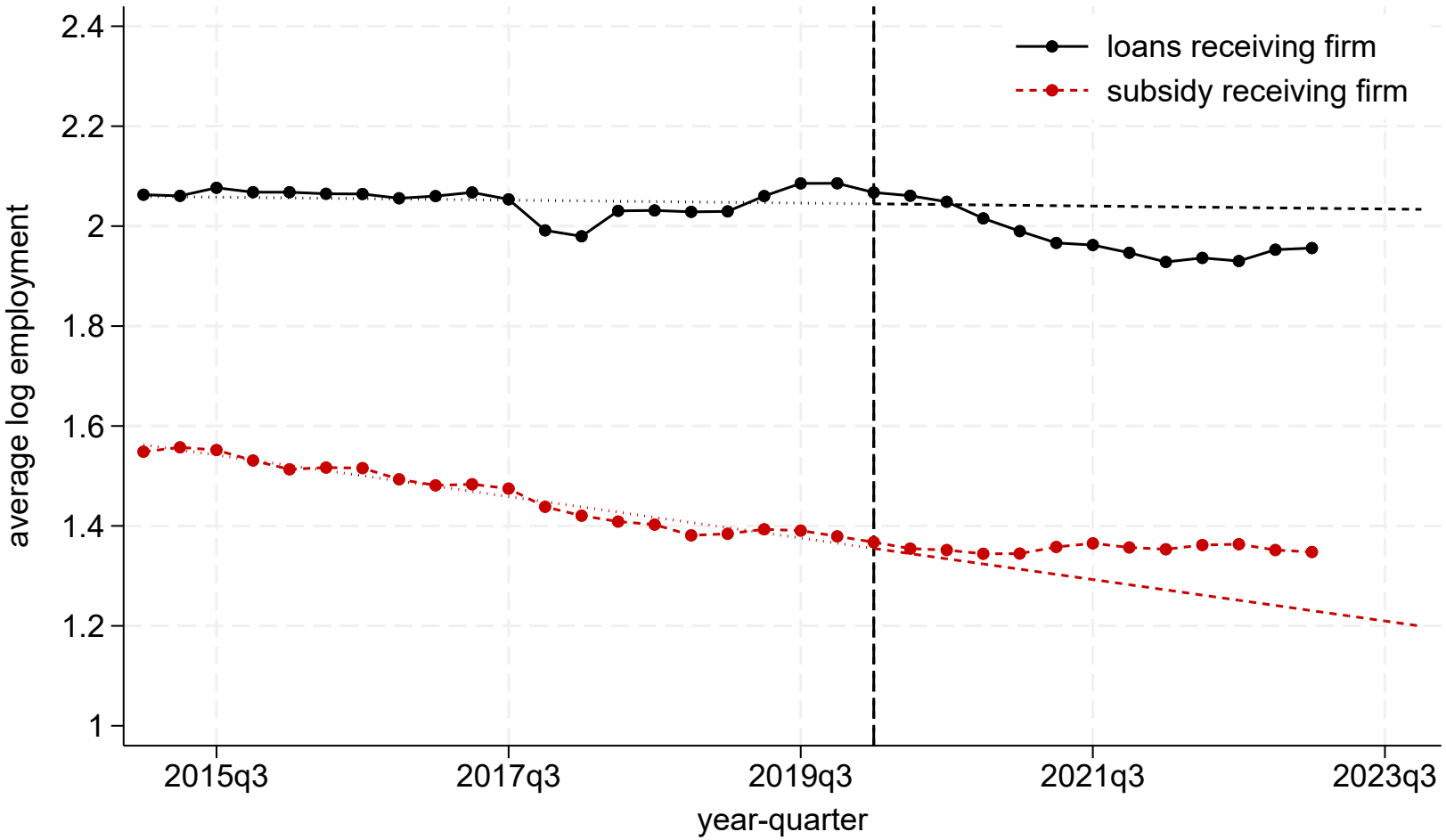
Employment level (COVID-19 support)



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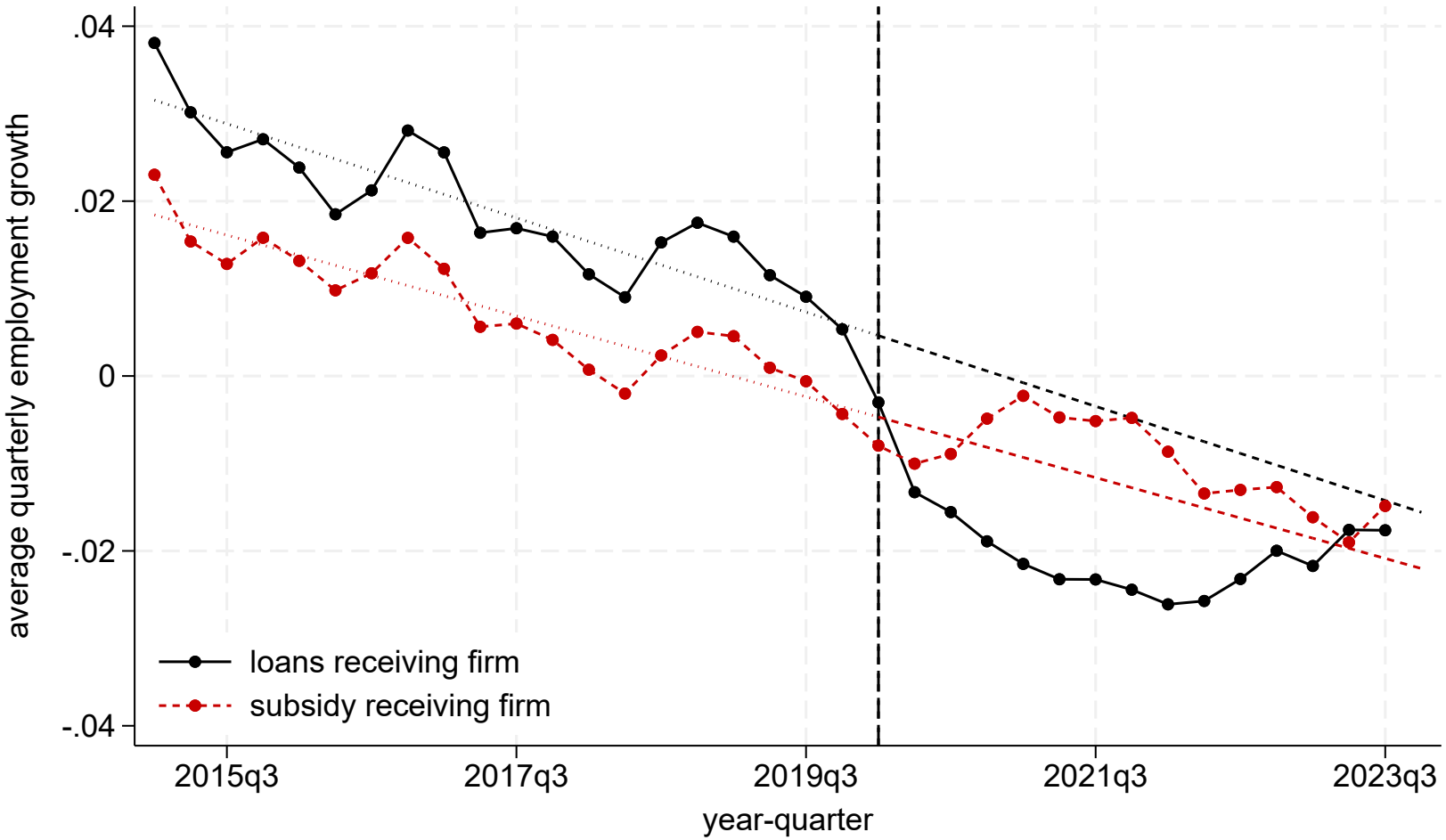
Employment growth (COVID-19 support)



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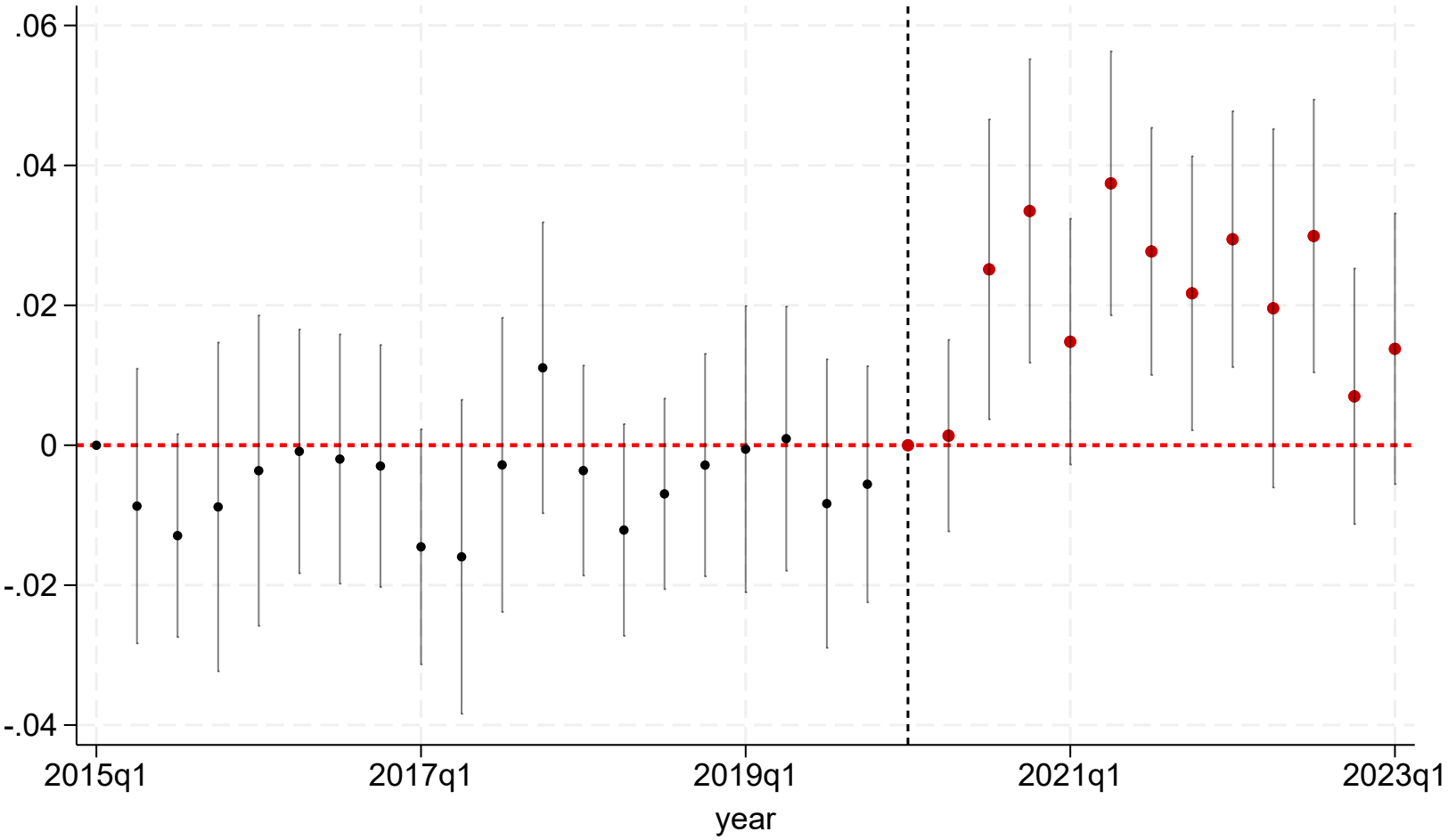
Relative employment growth of subsidy-receiving firms (COVID-19 support)



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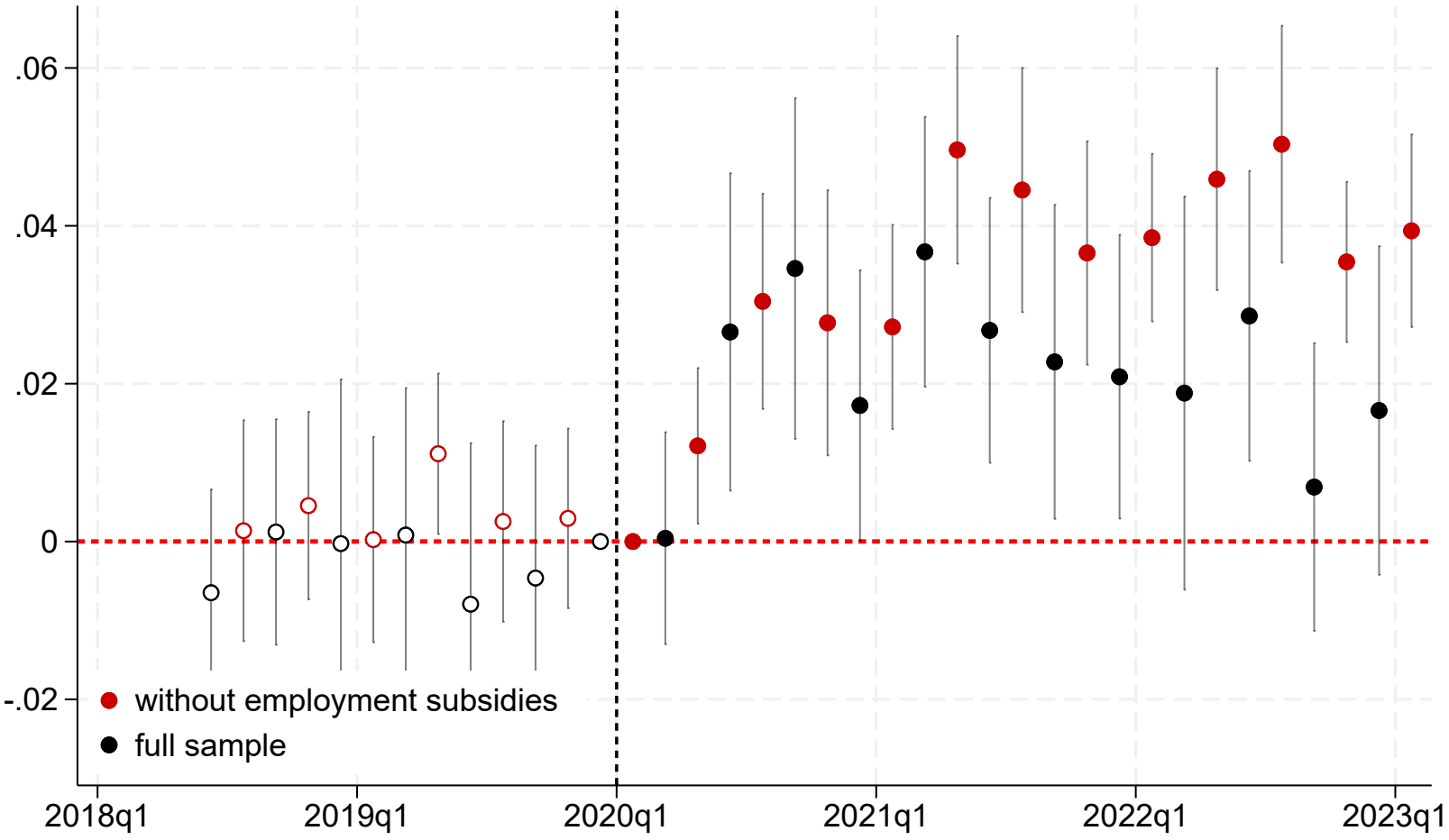
Relative employment growth of subsidy-receiving firms (COVID-19 support)



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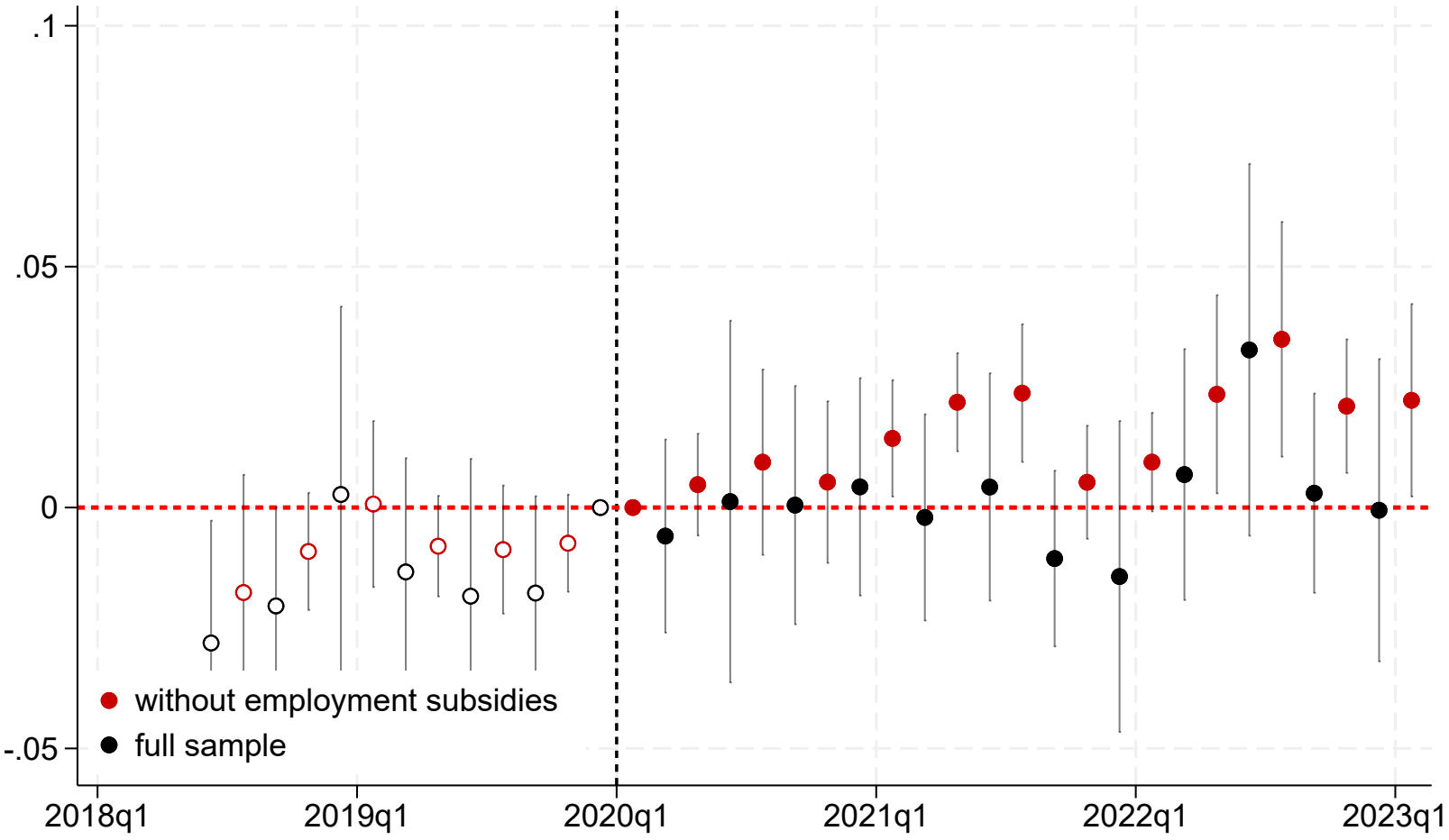
Relative employment growth of subsidy-receiving firms (COVID-19 support): without taxes



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Determinants of the Firm's Choice: Excluding Large Firms

Dep. var.:	Type of support indicator			
Base outcome:	No support			Subsidies only
Other outcome(s):	subsidies only	subsidies + loans	loans only	loans only
	(1)	(2)	(3)	(4)
log employment	0.163***	0.535***	0.984***	0.951***
	(0.011)	(0.016)	(0.031)	(0.036)
log total assets	-0.135***	-0.013	-0.038	0.109**
	(0.014)	(0.022)	(0.045)	(0.052)
log current assets ratio	-0.036***	-0.065***	0.031	0.105***
	(0.010)	(0.015)	(0.032)	(0.036)
log labor productivity	0.198***	0.235***	0.215***	0.040
	(0.007)	(0.012)	(0.024)	(0.028)
Observations	85,564			31,138

Back

Determinants of the Firm's Choice: Quarantine-Affected Firms

Dep. var.:	Type of support indicator			
Base outcome:	No support			Subsidies only
Other outcome(s):	subsidies only	subsidies + loans	loans only	loans only
	(1)	(2)	(3)	(4)
log employment	0.154***	0.577***	0.922***	0.890***
	(0.017)	(0.023)	(0.048)	(0.054)
log total assets	-0.064***	0.102***	0.157**	0.248***
	(0.020)	(0.030)	(0.070)	(0.080)
log current assets ratio	-0.012	-0.029	0.075	0.104*
	(0.014)	(0.020)	(0.050)	(0.054)
log labor productivity	0.180***	0.214***	0.180***	0.020
	(0.010)	(0.015)	(0.036)	(0.041)
Observations	39,888			17,592

Back

Determinants of the Firm's Choice: Revenue Fall of $\geq$ 30%

Dep. var.:	Type of support indicator			
Base outcome:	No support			Subsidies only
Other outcome(s):	subsidies only	subsidies + loans	loans only	loans only
	(1)	(2)	(3)	(4)
log employment	0.126***	0.692***	1.030***	0.985***
	(0.026)	(0.032)	(0.060)	(0.066)
log total assets	-0.406***	-0.380***	-0.234**	0.163
	(0.036)	(0.046)	(0.094)	(0.101)
log current assets ratio	-0.071***	-0.199***	-0.072	0.013
	(0.025)	(0.030)	(0.063)	(0.066)
log labor productivity	0.107***	0.205***	0.175***	0.088
	(0.018)	(0.024)	(0.051)	(0.054)
Observations	21,423			9,813

Back

Determinants of the Firm's Choice: Excluding Tax Exemptions and Deferrals, Keeping Quarantine-Affected Firms

Dep. var.:	Type of support indicator			
Base outcome:	No support			Subsidies only
Other outcome(s):	subsidies only	subsidies + loans	loans only	loans only
	(1)	(2)	(3)	(4)
lempl	0.147***	0.523***	0.732***	0.686***
	(0.016)	(0.033)	(0.098)	(0.109)
la_tot	-0.070***	0.169***	0.189	0.219
	(0.019)	(0.047)	(0.155)	(0.173)
lc_ratio	-0.007	-0.044	-0.065	-0.070
	(0.013)	(0.030)	(0.103)	(0.110)
llprod	0.170***	0.299***	0.271***	0.192**
	(0.009)	(0.024)	(0.076)	(0.085)
Observations	39,888			20,567

Back

Determinants of the Firm's Choice: Excluding Tax Exemptions and Deferrals, with Revenue Fall of $\geq$ 30%

Dep. var.:	Type of support indicator			
Base outcome:	No support			Subsidies only
Other outcome(s):	subsidies only	subsidies + loans	loans only	loans only
	(1)	(2)	(3)	(4)
lempl	0.151***	0.670***	0.953***	0.850***
	(0.024)	(0.042)	(0.104)	(0.107)
la_tot	-0.429***	-0.307***	-0.412**	-0.061
	(0.034)	(0.063)	(0.167)	(0.173)
lc_ratio	-0.085***	-0.171***	-0.064	0.066
	(0.023)	(0.040)	(0.111)	(0.115)
llprod	0.099***	0.338***	0.363***	0.326***
	(0.017)	(0.033)	(0.089)	(0.092)
Observations	21,423			11,939

Back